

Rockstone — User Manual

A hands-on manual for the Rockstone desktop application. Rockstone processes rock-sample images — especially micro-CT stacks — with **slice alignment**, **beam-hardening removal**, **background correction**, view/measurement tools and multi-format export.

For build/development notes see `README.md`; this document is written for *users* of the app.

User manual — generated from `MANUAL.md`

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1. Getting started

Launch the app

- Double-click `Rockstone.bat` (Windows), or run:

```
rockstone gui
```

or

```
python -m rockstone.gui
```

- You can pass a file to open at startup: `rockstone gui myfile.tif`.

Install (first time)

```
pip install -e .
```

Requires Python 3.10+ with numpy, scipy, tiffile, imageio, pydicom and Pillow (installed automatically).

2. Loading data

Open file... (Ctrl+O) opens one data file. In the file dialog you can Ctrl/Shift-click to select **several image files at once** — they are stacked into one volume in natural (numeric) order.

Open folder... (Ctrl+Shift+O) loads every slice image inside a folder (subfolders included) or a **DICOM series** (all slices of one series in a folder, sorted by position).

What happens on load:

- the **info panel** (top of the side panel) shows file name, data

kind, shape, value range, cupping severity and the physical voxel size if known;

- the first slice is displayed; browse with the **Slice** slider;
- the title bar shows `Rockstone – rock sample processing` and gains a

* whenever there are unsaved changes.

Data kinds. Rockstone distinguishes *volumes* (reconstructed stacks, the usual case) from *projections* (raw detector frames). The kind is auto-detected and shown in the info panel; it decides which corrections are offered (e.g. "Linearize projections (chord)" instead of "Remove beam hardening" for raw data).

3. The main window

```
+-----+
| Menu bar: File Edit View Process Help |
+-----+
| Toolbar: open | process | compare | zoom | view | save |
+-----+-----+
| | info panel |
| slice image | [Appearance] |
| (wheel = zoom, drag = pan) | colors |
| | brightness |
| | contrast |
| | histogram |
| | [Export & |
| | compare] |
| | [Slice] |
| | slider |
+-----+-----+
| status bar |
+-----+
```

- **Menu bar** — every command with its shortcut (see §12).
- **Toolbar** — the most-used commands, grouped by separators.
- **Side panel** — *Appearance* (color system, brightness/contrast,

histogram), *Export & compare*, *Slice* navigation, and the info read-out on top.

- **Status bar** — progress and result messages ("alignment finished.", "cropped to 128 x 128 px — 24 slices kept", ...).

4. Viewing: zoom, pan, slices, colors

Zoom and pan

- **Mouse wheel** over the image zooms in/out (10%–2000%).
- **Click-drag** moves (pans) the image.
- Toolbar – **+** / **-** buttons, the **View** menu, or **Ctrl+–** / **Ctrl++**.

- **Fit** (Ctrl+F) scales the slice to the window.
- Zoom and pan survive slice scrolling and color changes; loading a new file resets to 100%.

Slices

- Drag the vertical **Slice** slider, or pick a thumbnail in the

slice map (§11). The label under the slider shows `current / total`.

Color systems

The **Colors** dropdown re-renders the slice through a lookup table:

Name	Good for
Gray	default, neutral
Inverted	pores and dark detail
Hot metal / Viridis / Jet	density gradients
Rock	natural rock look
Asphalt	cool neutral grays (pavement)
Ice / Copper / Thermal / Bone	stylistic renders

Changing colors is display-only and instantly re-renders the slice map too.

5. Brightness, contrast and the histogram

The **Brightness** and **Contrast** sliders (−100...100) adjust only the *display window* — pixel values, the histogram of the data and saved files are untouched.

- **Brightness** slides the window up/down.
- **Contrast** narrows the window (positive) or widens it (negative).
- **Reset display** returns to the automatic 1–99.5 percentile stretch

(also in the View menu).

The **histogram panel** shows the intensity distribution of the currently displayed image. It follows slicing, color systems and the display window — handy for judging whether a window choice saturates the sample.

6. Processing: align, dehardening, background

All processing runs in the background (the window stays responsive), is **undoable** (§9) and marks the data as changed.

Align slices (Process ■ Align slices)

Removes slice-to-slice jitter (translation + in-plane rotation) using phase correlation with Fourier–Mellin rotation estimation. Works for volumes and raw projections. The status bar reports progress; the info panel's cupping value is re-computed afterwards.

Remove beam hardening (Process ■ Remove beam hardening)

Fixes **cupping** — dense material near the edge reading brighter than in the center — with a radial multiplicative gain field (volumes), or linearizes raw projections against chord length. Compare the cupping value in the info panel before/after to judge the effect.

Background correction... (Process ■ Background correction...)

Flattens the level and shading *around* the sample (detector offset, reconstruction pedestal, scattering halos):

- **Mode** — *Subtract* (air becomes 0; usual for CT stacks) or

Divide by shading (multiplicative shading flattens to a constant).

- **Estimate per slice** — on by default; untick for one shared

surface.

- **Background** — *dark* (air around bright rock) or *bright*.
- **Air percentile** — the percentile used as the air level (default

1.0).

Afterwards the status bar reports e.g. `air 0.13 -> 0.0008, bg spread 0.053 -> 0.005`.

Recommended order: Align ■ Remove beam hardening ■ Background correction.

7. Rotate, crop, ROI and physical dimensions

Measure ROI... (Edit menu or ROI... button)

Measures a rectangular **region of interest** on the raw data:

1. Click **ROI...** and **drag a rectangle** on the image (cyan outline;

the red outline is the crop tool).

1. The dialog shows **mean, std, min/max, median, pixel count** and

the **area in mm²** (when physical dimensions are set), plus a mean-above-air statistic referenced to the standard CT air value of **-1000 HU**.

1. Scroll slices — the statistics **update live** for the same region.

- **Crop to ROI...** converts the measurement into an undoable crop.
- **Save all slices (CSV)...** writes one row per slice with the same

statistics — perfect for tracking a feature through the depth of the sample.

Rotate 90° (Edit menu, ■/■ toolbar buttons)

Rotates the **whole stack** in-plane; slice count and order are unchanged. Undoable like every other modification.

Crop... (Edit menu or toolbar)

1. Click **Crop...** — a small dialog appears and the cursor becomes a

crosshair.

1. Choose the **Shape**:

- **Rectangle** — crops the stack down to the dragged selection;
- **Ellipse** — keeps the ellipse inscribed in the dragged

selection on every slice and fills the outside with **NaN** (transparent — excluded from statistics and exports). The image size is unchanged, so slices stay registered — perfect for cutting the surrounding material away from a cylindrical sample.

1. **Drag the region** on the image. The outline follows the chosen

shape at any zoom; the dialog shows its size and position in pixels.

1. Press **Apply** to keep that region in every slice, or

Cancel / Escape to abort.

Cropping never changes the slice count, the data kind or the stored physical dimensions. Undoable.

Physical dimensions... (Edit menu or Dimensions... button)

Sets the voxel size in millimeters — pixel width, pixel height and slice spacing. The values:

- appear in the info panel (voxel: 0.05 x 0.05 x 0.05 mm);
- are **embedded when saving** TIFF/NPZ, so processed files keep their

calibration (`rockstone.read_metadata` reads them back);

- are prefilled from DICOM input (`PixelSpacing`, `SliceThickness`) or

from metadata embedded in the loaded file.

Clearing all fields removes the dimensions again.

8. Compare and difference view

Compare... (View menu or toolbar) asks for a second dataset (e.g. the original file while the processed one is open) and shows:

- **side-by-side** view of the same slice of both stacks, then
- **Show difference** toggles an intensity-difference image.

The status bar reports difference statistics (max/mean/RMSE, PSNR, correlation). Press **Escape** or click **Show difference** again to return to the normal view.

9. Undo, redo and saving

- **Undo** (Ctrl+Z) / **Redo** (Ctrl+Y or Ctrl+Shift+Z) step through

the last **10 modifications** (align, deharden, background, rotate, crop). A new modification drops the redo fork, as in standard editors.

- The unsaved-changes * in the title bar is identity-based: undoing

back to the last saved state clears it automatically.

Saving

- **Save** (Ctrl+S) writes back to the file the data came from

(TIFF/NPY/NPZ).

- **Save As...** (Ctrl+Shift+S) asks for a destination; after a

correction the suggestion is `<name>_processed.tif`.

- Data that came from a folder or a multi-file selection always falls

back to Save As.

- Closing the app or loading another file with unsaved changes asks

whether to **save first** (the save there is synchronous, so it is safe to re-read the same file).

- Embedded physical dimensions (if set) are written into the file.

10. Exporting images and data

Report... (side panel, Process ■ Generate report..., Ctrl+R)

Writes a **summary report** of the current dataset as **HTML** (with embedded snapshot and histogram — no external files), **PDF**, **Word (.docx)** (both with the same images) or **plain text**. Contents: shape, kind, value range, physical dimensions and sample extent in mm, cupping / air level / background-spread metrics with hints, the modification history of the session, compare results, and a note when there are unsaved changes. The

suggested name is `<file>_report.html` (extensions follow the chosen format). Headlessly: `rockstone report scan.tif -o report.html (--pdf, --docx, --txt, --no-image)`.

Export... (side panel, Process ■ Export current slice..., Ctrl+E)

Saves the **current slice**:

Format	What you get
PNG	the rendered view as displayed (colors + window), tagged with the physical pixel size as DPI so 1:1 prints are true scale
SVG	<i>true vector</i> : one shape per run of equal pixels, canvas sized in millimeters from the voxel metadata, raw value range recorded
PDF	the rendered view as a one-page document, sized by the physical DPI
CSV	the <i>raw</i> (float) values of the slice as <code>row</code> , <code>col</code> , <code>value</code> rows; NaN → empty cell

The save dialog suggests `<file>_slice<n>.<ext>` with the current slice number. Export runs in the background; the status bar confirms.

Note: PNG/PDF/SVG contain the *rendered* image (what you see, including brightness/contrast); CSV contains the *data*. For publications prefer SVG (infinite zoom, exact physical scale) plus CSV for the numbers.

11. The slice map

Slice map (toolbar button, or View ■ Slice map, Ctrl+M) opens a separate scrollable window showing **every slice at once**. Click a thumbnail to jump to it; a red rectangle marks the current slice and follows as you scroll. The map rebuilds automatically after any data or color change and closes with the main window. Toggle it closed with the same button or by closing the map window.

12. Keyboard shortcuts

Keys	Action
Ctrl+O	Open file...
Ctrl+Shift+O	Open folder...
Ctrl+S	Save
Ctrl+Shift+S	Save As...
Ctrl+E	Export current slice...
Ctrl+Z	Undo
Ctrl+Y / Ctrl+Shift+Z	Redo
Ctrl+M	Slice map on/off
Ctrl+R	Generate report...
Ctrl++ / Ctrl+–	Zoom in / out
Ctrl+F	Fit to window
Escape	Cancel crop / leave compare view
mouse wheel	Zoom
click-drag	Pan (or draw the crop/ROI rectangle in crop/ROI mode)

13. The command line

All processing is also available headlessly:

```
rockstone info data.tif # shape, dtype, kind, range, cupping
```

```

rockstone align raw.tif -o a.tif # rigid slice alignment
rockstone deharden vol.tif -o d.tif # cupping removal
--method radial|chord|poly --degree 3 --background dark|bright
rockstone background vol.tif -o b.tif # air-level / shading flattening
--mode offset|divide --shared --percentile 1.0 --background dark
rockstone run scan.tif -o out.tif # full pipeline (align + deharden)
--with-background # + background correction
--no-align --no-deharden # stage toggles
rockstone compare a.tif b.tif # difference report
rockstone report scan.tif -o r.html # dataset report (HTML; --txt, --no-image)
rockstone roi scan.tif --rect 100 100 199 199 --slice 5
# ROI statistics (--all-slices, --csv)
rockstone demo # generate a synthetic demo dataset

```

Every command accepts `-o` (output file) and `-p` (per-slice progress lines). Default output is `<input>_processed.tif`.

14. Supported file formats

Source	Details
TIFF	single- or multi-page stacks (no JPEG-compressed TIFF)
DICOM	single files, multi-frame files, or series folders (z-sorted, HU rescale applied); <code>PixelSpacing/SliceThickness</code> become the physical dimensions
NPY / NPZ	raw NumPy arrays (NPZ as saved by Rockstone)
Image folders	PNG/JPEG/TIFF slices, subfolders included, natural numeric order
Output	TIFF (with metadata), NPY, NPZ; slice export: PNG, SVG, PDF, CSV

2-D, 3-D and multichannel (RGB) data are handled; the stack convention is `(slices, rows, columns)`.

15. Tips and troubleshooting

The correction made things worse? Undo (Ctrl+Z) and check the metrics: cupping (info panel) for beam hardening, air level/background spread (status bar) for background correction. The numbers are printed before/after on purpose.

The sample fill the whole frame? Crop first (§7) — the background correction then has clean air to fit, and exports are tighter.

SVG export is large. The SVG contains one shape per pixel run; on noisy 1024² slices this can be big. Export a **PNG** for previews and SVG for final figures (or after smoothing).

DICOM series loads with wrong slice order? The series is sorted by the DICOM z-position; if the headers are broken, sort the files into a folder with numbered names and open the folder instead.

"unsupported output format" — Save As only offers `.tif/.tiff/.npy/.npz`; use **Export...** for PNG/SVG/PDF/CSV.

JPEG-compressed TIFF won't open — the JPEG codec (imagecodecs) is not installed; convert the file, or `pip install imagecodecs` and reload.

Nothing happens when clicking a button — the status bar explains: during background tasks most controls are disabled until the task finishes.

Data looks all black / all white — the display is percentile stretched; press **Reset display**, or use the brightness/contrast sliders. The raw values are never modified by display controls.

Rockstone is under active development; see [README.md](#) for the feature list, methods and the test suite.